

# Dr. Tom Lauwaerts

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Postdoctoral researcher passionate about programming language design and debuggers.

## Work experience

### Post-doctoral Researcher

Vrije Universiteit Brussel · 2025 - present

### PhD Researcher: Programming Languages

Ghent University · 2021 - 2025

Research into debugging and testing tools with a strong formal foundation.

### Teaching Assistant

Ghent University · 2021 - 2025

Courses: *Fundamenten van programmeertalen*, *Logisch programmeren*.

### Embedded Software Developer (Internship)

Creative Therapy (startup) · July - August 2020

## Education

### PhD in Computer Science · 2021 - 2025

Ghent University, TOPL Lab

### BSc. and MSc. in Computer Science · 2016 - 2021

Ghent University, Faculty of Sciences

## Highlighted research



### MIO: Multiverse Debugging in the Face of Input/Output **FULL PAPER**

2025-10 | **Tom Lauwaerts**, Maarten Steevens, Christophe Scholliers | *Proceedings of the ACM on Programming Languages (PACMPL)*, Volume 9, Issue OOPSLA2

## Academic services

### IC00OLPS workshop co-organizer · 2026 - 2027

Co-organizing the IC00OLPS'26 and IC00OLPS'27 workshops.

### ECOOP student volunteer chair and local organizer · 2026

Part of the VUB team organizing ECOOP'26 in Brussels.

### DEBT workshop co-organizer · 2025

Co-organizing the DEBT'25 workshop.

### Programming artifact evaluation committee member · 2024 - 2027

AEC member for the Programming conference and journal (vol. 9 to 12).

## Additional information

### Science communication and outreach · 2021 - present

Alumniday UGent, Dag van de Wetenschap, Comp Sci Olympiad ...

### Languages

**Dutch** native · **English** high proficiency · **French** elementary · **Chinese** novice

## Awards

### ISSTA'23 Research competition: 3rd place · 2023

ACM Student Research Competition at ECOOP/ISSTA 2023 3rd place medal.

**Concolic Multiverse Debugging**  FULL PAPER

2026-06 | Maarten Steevens, **Tom Lauwaerts**, Christophe Scholliers | *Special Issue of the 40th European Conference on Object-Oriented Programming (ECOOP 2026)*

**MIO: Multiverse Debugging in the Face of Input/Output**  FULL PAPER

2025-10 | **Tom Lauwaerts**, Maarten Steevens, Christophe Scholliers | *Proceedings of the ACM on Programming Languages (PACMPL), Volume 9, Issue OOPSLA2*

**Foundations for Constrained Debugging Techniques** DISSERTATION

2025-08 | **Tom Lauwaerts** | *PhD dissertation (Universiteit Gent)*

**Latch: Enabling large-scale automated testing on constrained systems** JOURNAL PAPER

2024-12 | **Tom Lauwaerts**, Stefan Marr, Christophe Scholliers | *Science of Computer Programming Journal*

**Concolic Multiverse Debugging** DEMO

2024-09 | Maarten Steevens, **Tom Lauwaerts**, Christophe Scholliers | *2nd ACM International Workshop on Future Debugging Techniques (DEBT 2024)*

**WARDuino: An embedded WebAssembly virtual machine** JOURNAL PAPER

2024-06 | **Tom Lauwaerts**, Robbert Gurdeep Singh, Christophe Scholliers | *Journal of Computer Languages*

**Out-of-Place Debugging on Constraint Devices with the EDWARD Debugger (Demo)** DEMO

2023-07 | **Tom Lauwaerts**, Carlos Rojas Castillo, Elisa Gonzalez Boix, Christophe Scholliers | *DEBT'23*

**Demo: Debugging Constraint Devices with EDWARD** DEMO

2023-06 | **Tom Lauwaerts**, Carlos Rojas Castillo, Elisa Gonzalez Boix, Christophe Scholliers | *MobiSys'23*

**Event-Based Out-of-Place Debugging** FULL PAPER

2022-09 | **Tom Lauwaerts**, Carlos Rojas Castillo, Robbert Gurdeep Singh, Matteo Marra, Christophe Scholliers, Elisa Gonzalez Boix | *MPLR'22*